Atomic Structure

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Protons are positive and in the nucleus, neutrons have no charge and are in the nucleus, and electrons are negative and outside the nucleus.

Round 1 - Retrieve It

1. What charge does a proton have, and where is it found?

2. What charge does a neutron have, and where is it found?

3. What charge does an electron have, and where is it found?

4. In a neutral atom, how do the numbers of protons and electrons compare?

Round 2 - Sketch It

Draw a simple model of a neutral carbon atom with 6 protons, 6 neutrons, and 6 electrons. Label the nucleus and electron region.

Atomic Structure - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. An atom has 8 protons, 8 neutrons, and 10 electrons. What is its overall charge? Explain.

2. Two atoms each have 6 protons. One has 6 neutrons and the other has 8. Are they the same element?

3. If an atom loses one electron, does its nucleus change? What happens to its charge?

Round 4 - Reason from Evidence

Explain why changing the number of neutrons can make a different isotope, while changing the number of protons makes a different element.

Atomic Structure - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate
Explain Atomic Structure in your own words. Use one example that is not printed on the cue card.

C

Connect
Connect atomic structure to the periodic table. Which particle determines an element's atomic number?

E

Extend
A mystery atom has 13 protons and 10 electrons. Predict its charge and explain what evidence you used.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Atomic Structure

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Positive; in the nucleus.

2. No charge; in the nucleus.

3. Negative; outside the nucleus in the electron cloud.

4. They are equal.

Sketch It - look for

A nucleus labeled with 6 protons and 6 neutrons, with 6 electrons shown outside the nucleus. Exact shell placement is not required unless taught.

Apply It

1. 2- because it has two more negative electrons than positive protons.

2. Yes. Element identity is set by the number of protons; these are isotopes of the same element.

3. The nucleus does not change; the atom becomes more positive by one charge.

Reason from Evidence - look for

Look for: element identity depends on proton number; neutron number can vary within the same element; changing protons changes atomic number and therefore the element.

ACE - Extend look-for

Atomic number comes from protons. Mystery atom charge is 3+ because 13 positive protons outweigh 10 negative electrons by three.

Suggested use	Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.